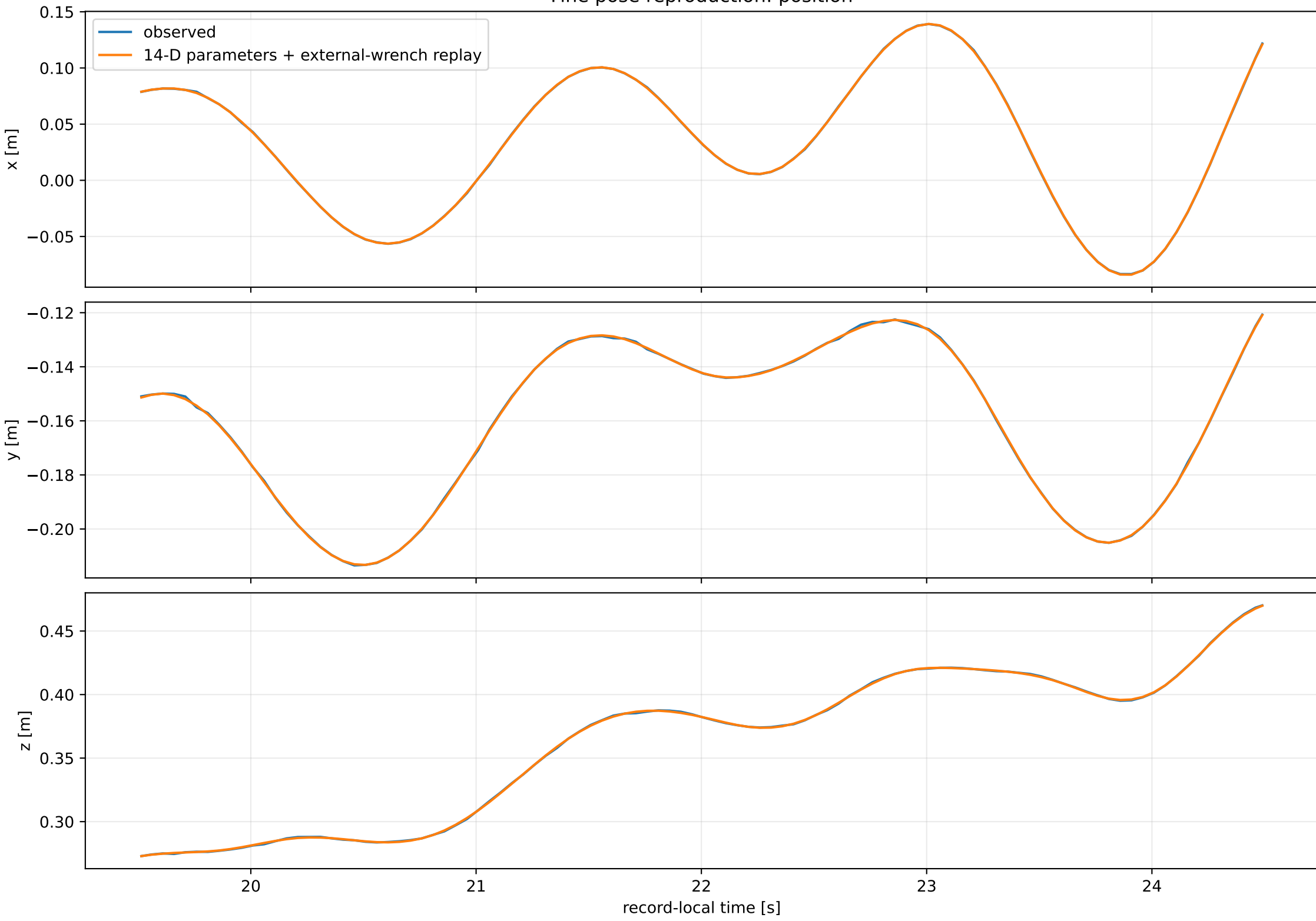
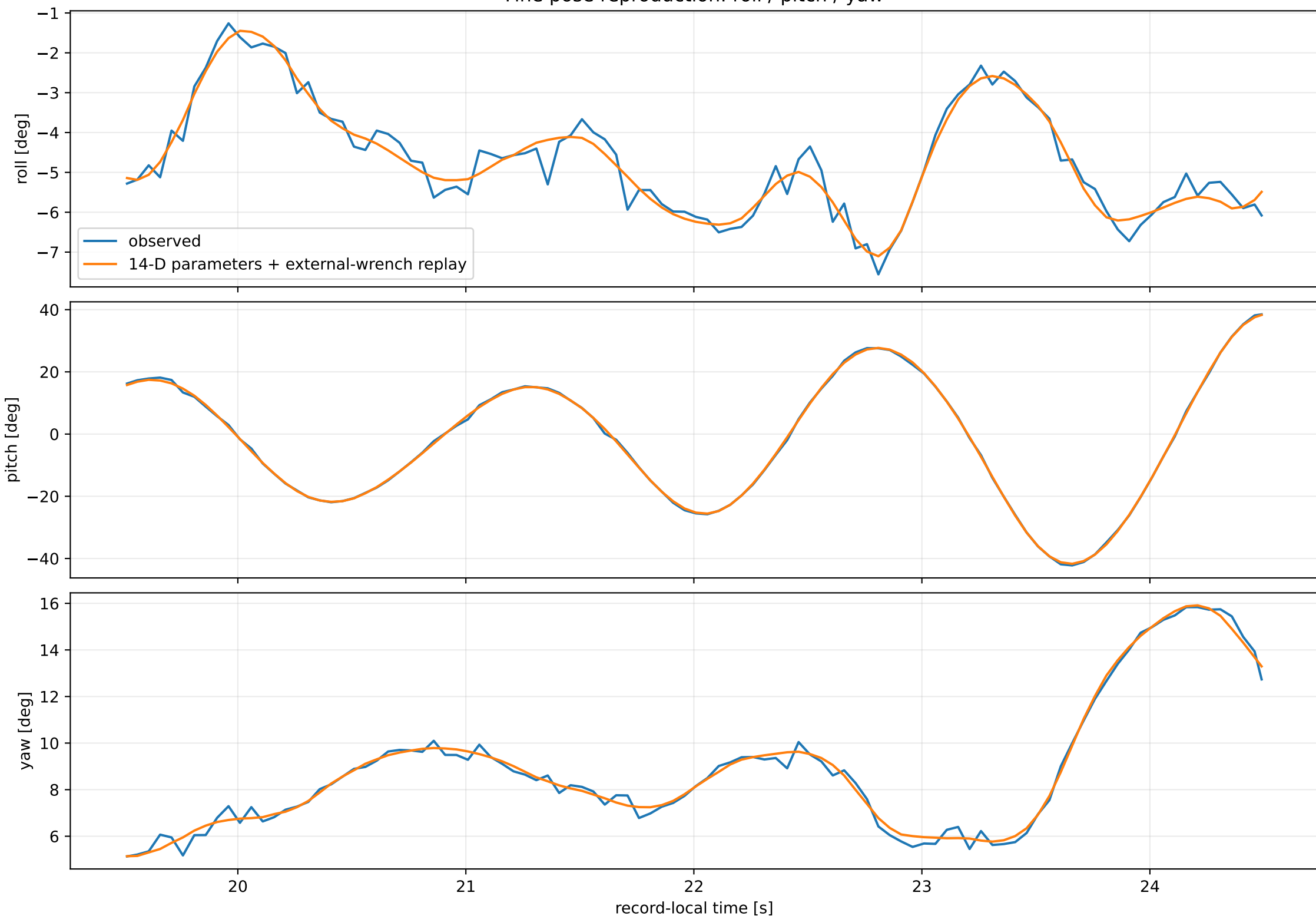


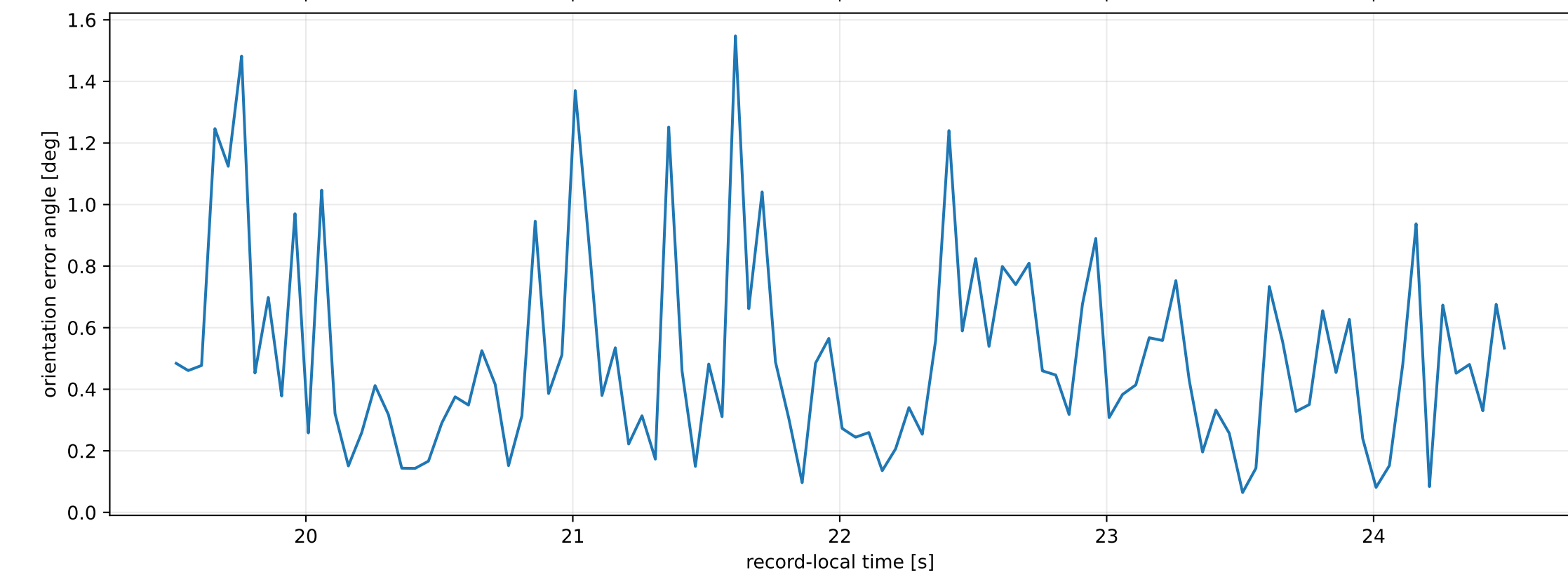
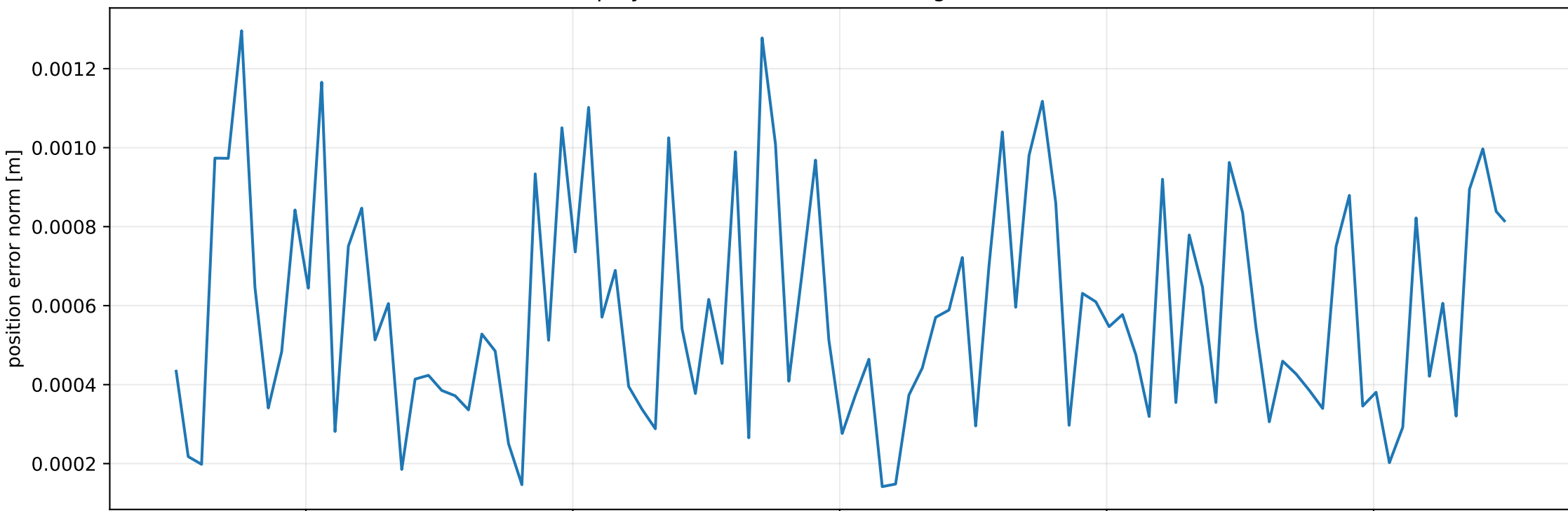
Fine pose reproduction: position



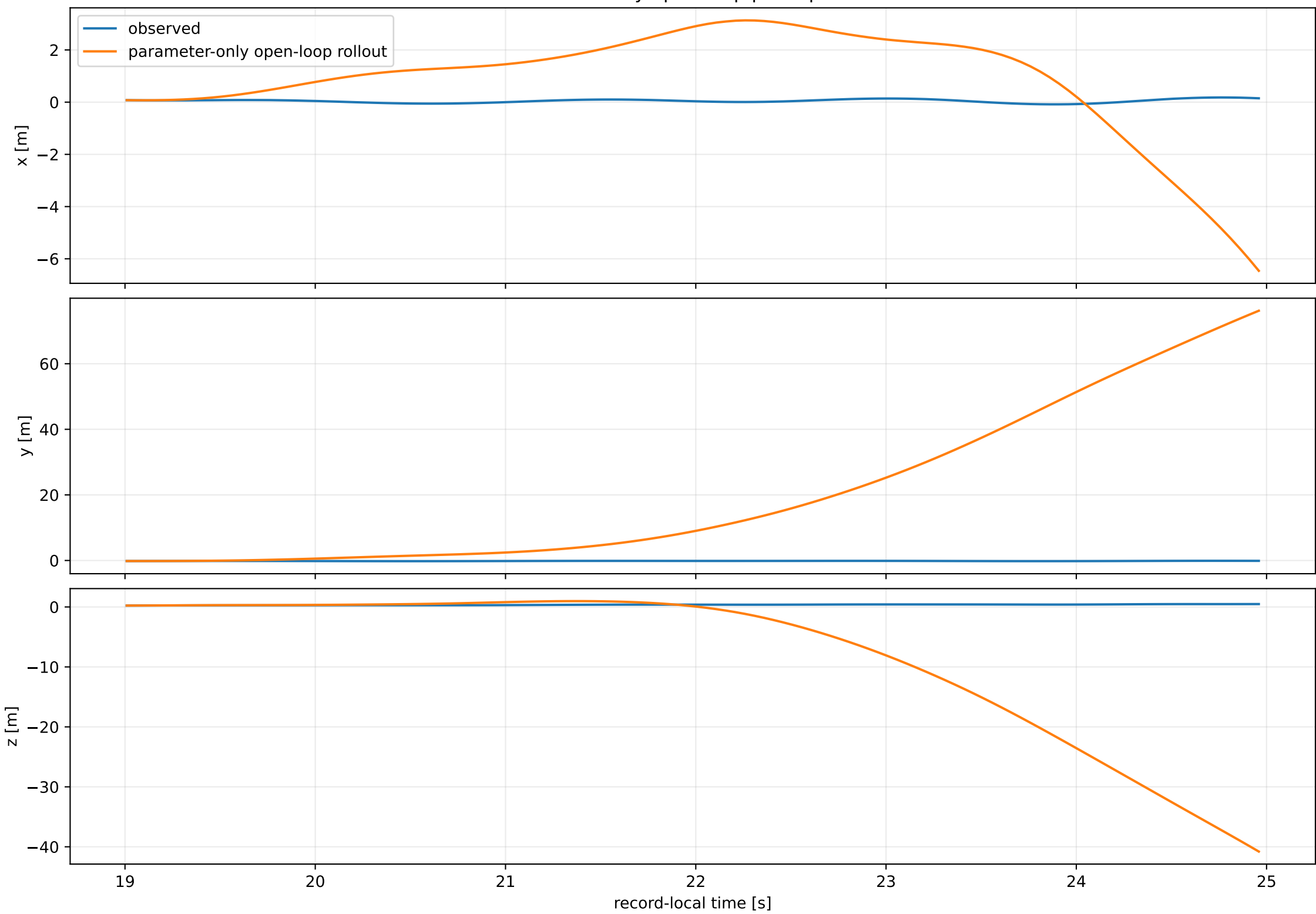
Fine pose reproduction: roll / pitch / yaw



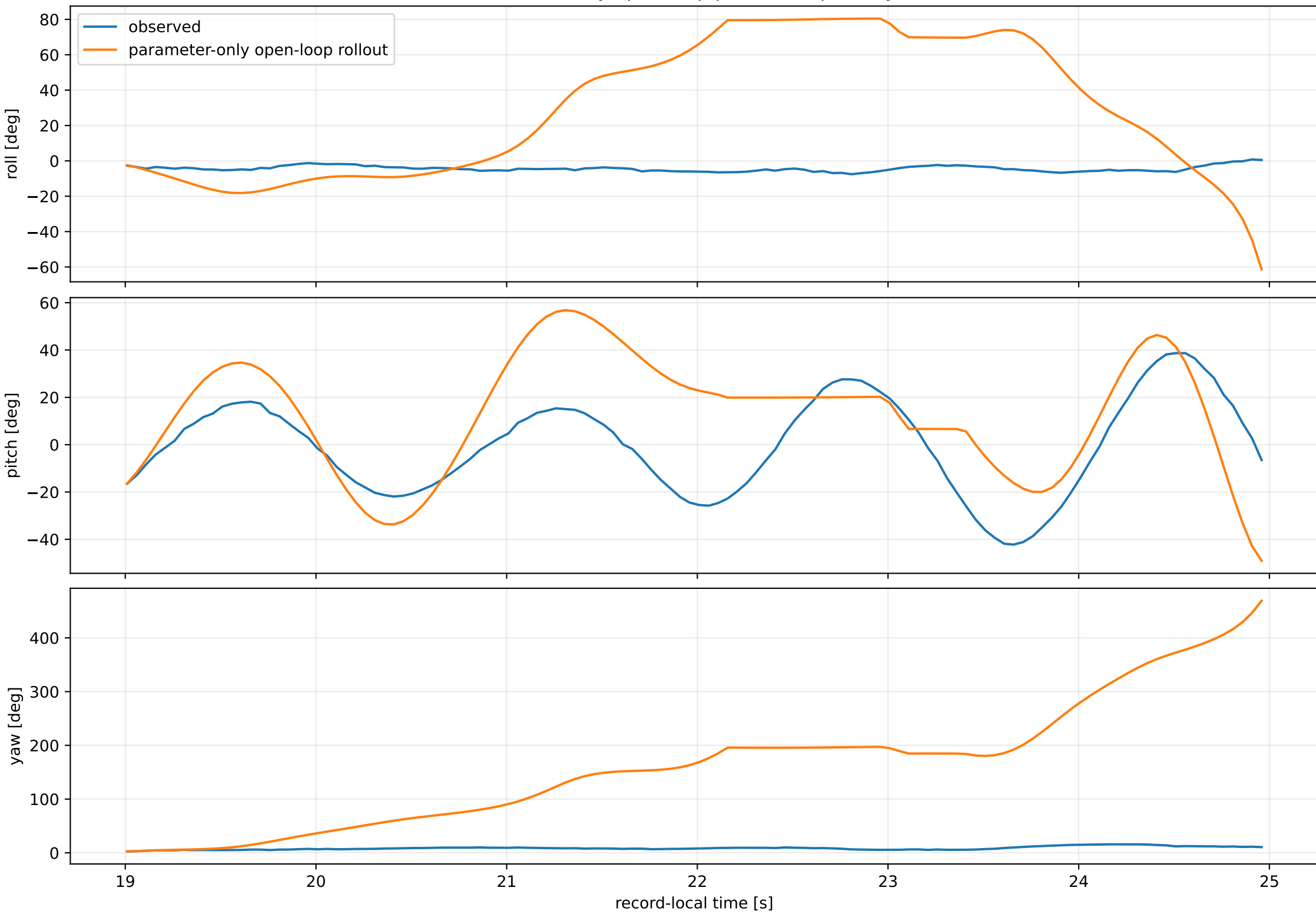
Fine replay error after the 14-D change — lower is better



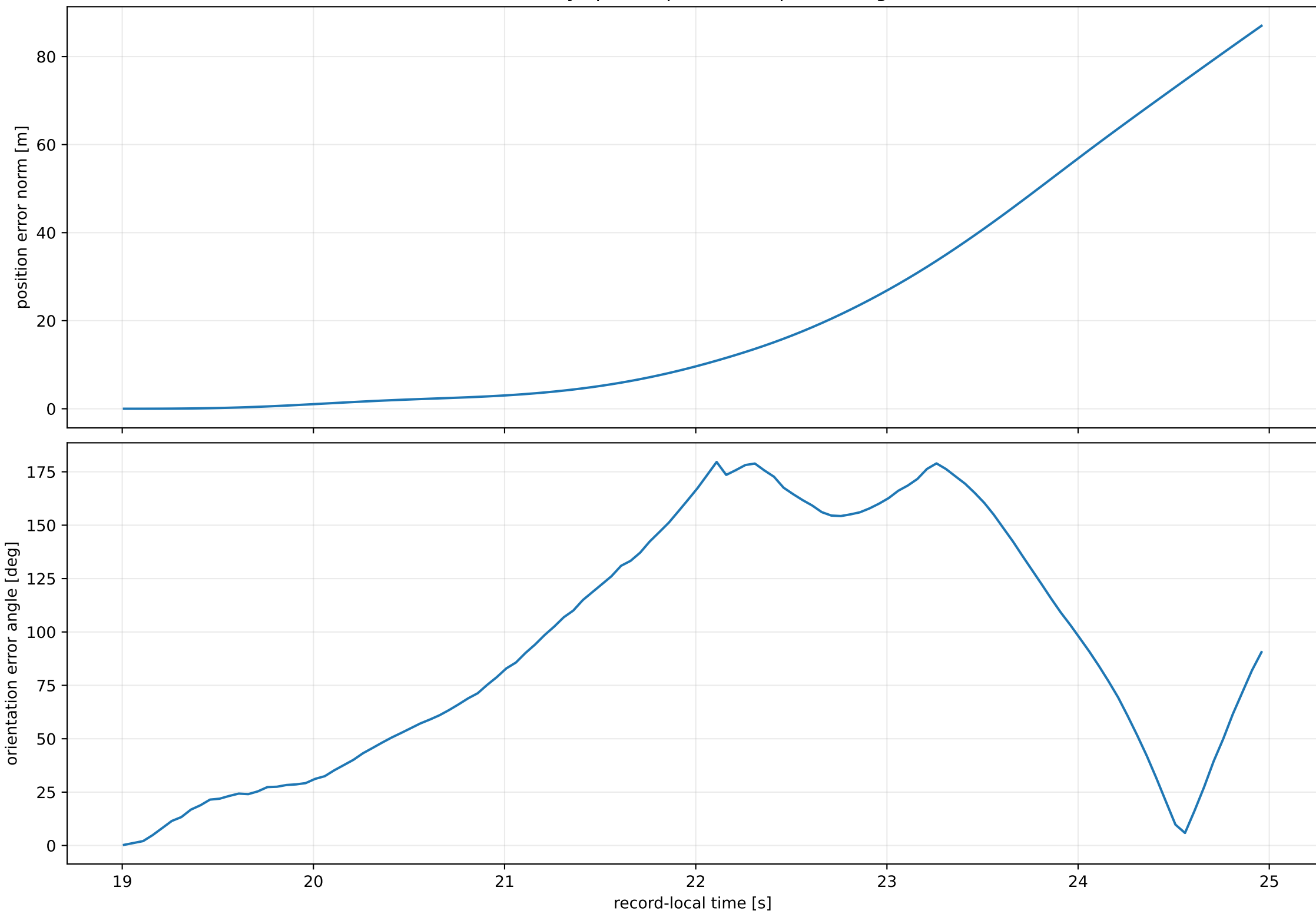
Parameter-only open-loop pose: position



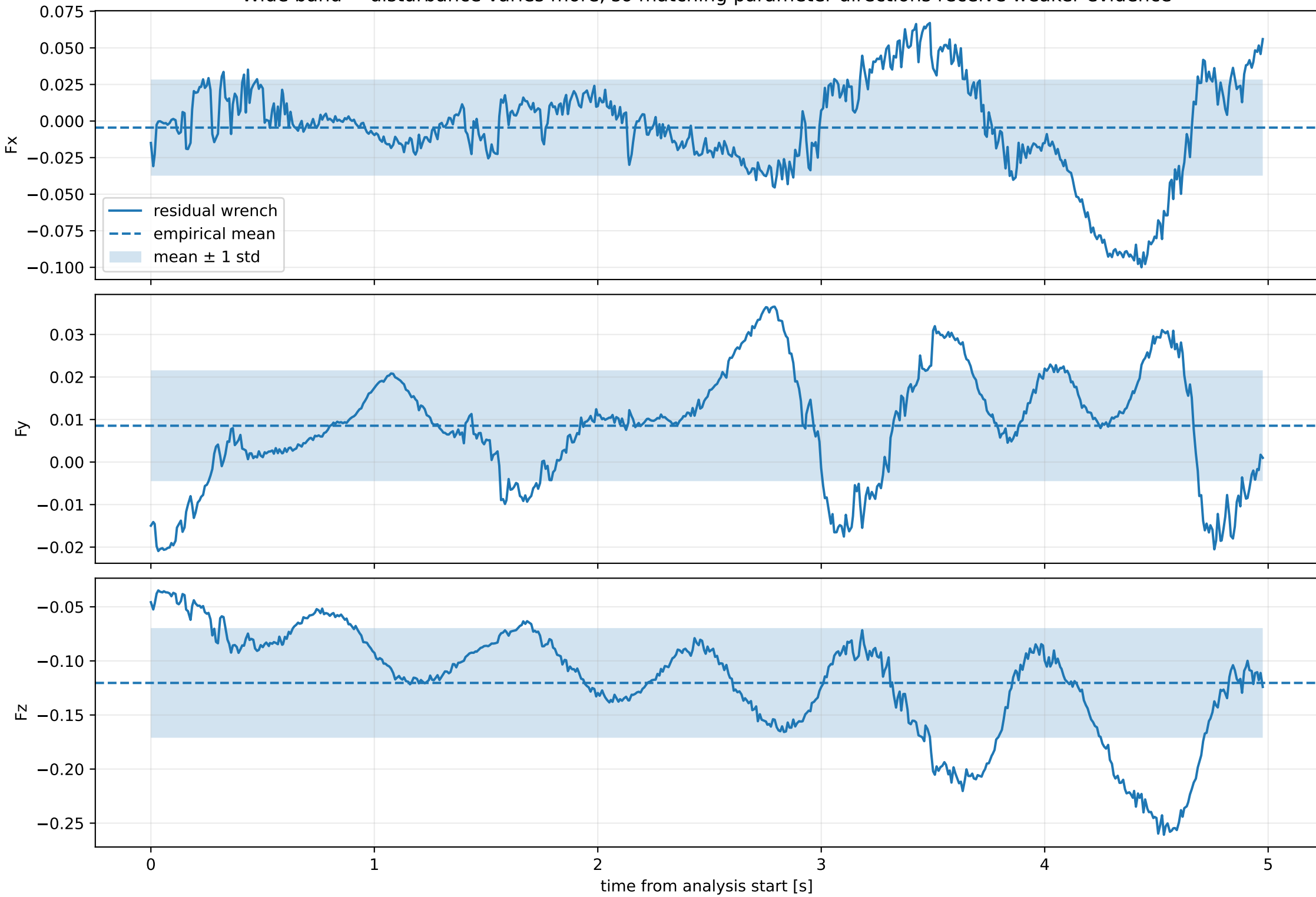
Parameter-only open-loop pose: roll / pitch / yaw



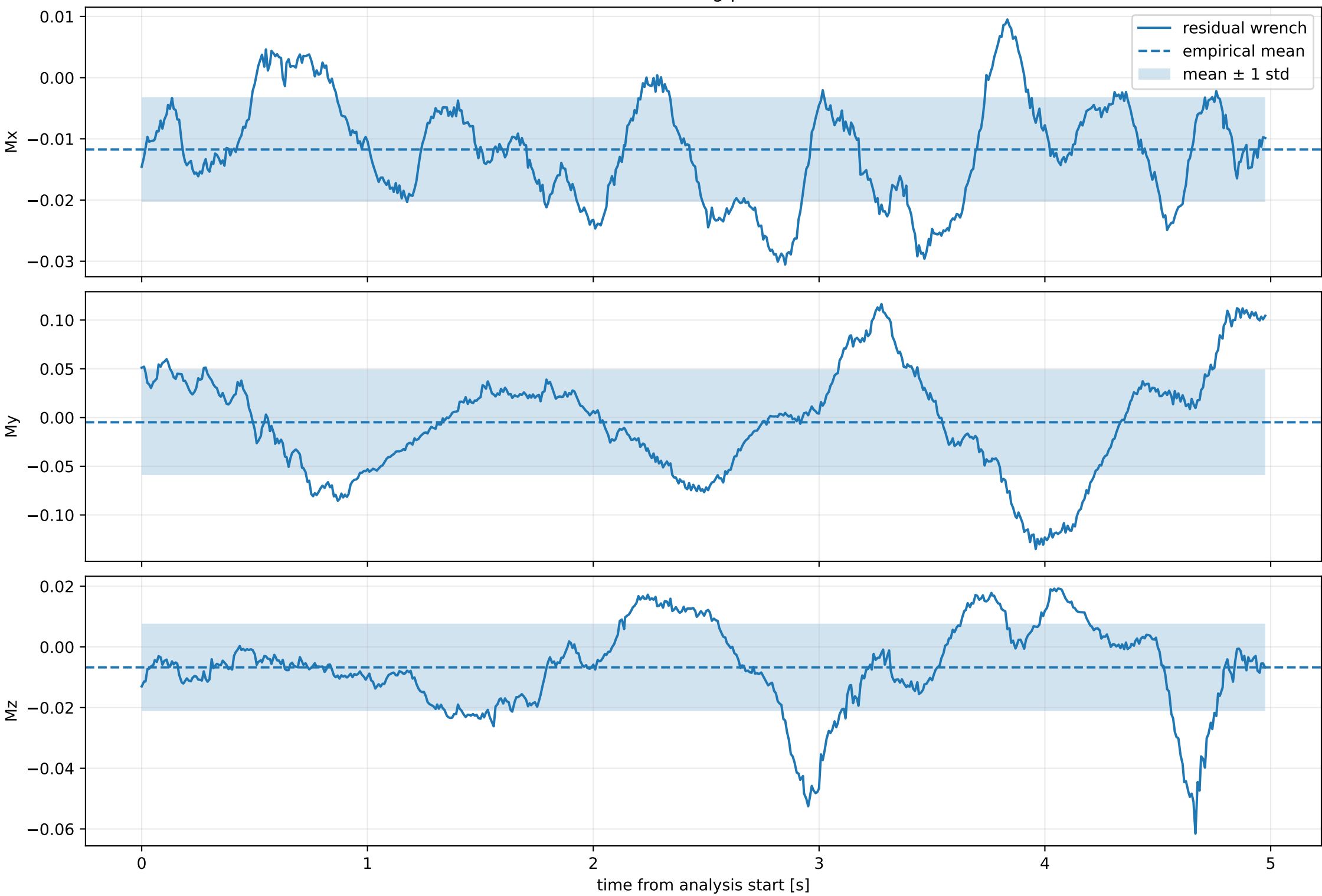
Parameter-only open-loop rollout (separate diagnostic)



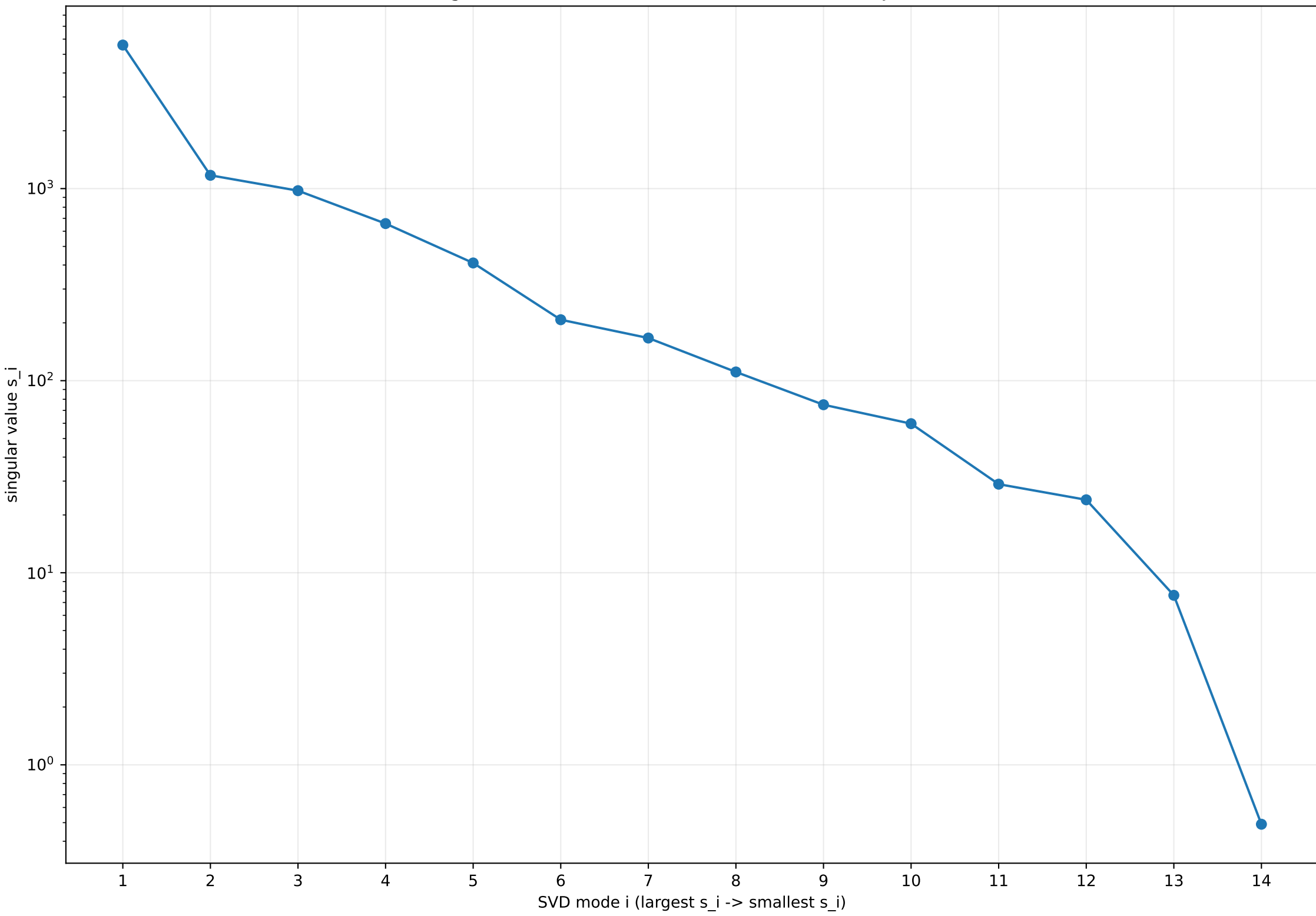
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



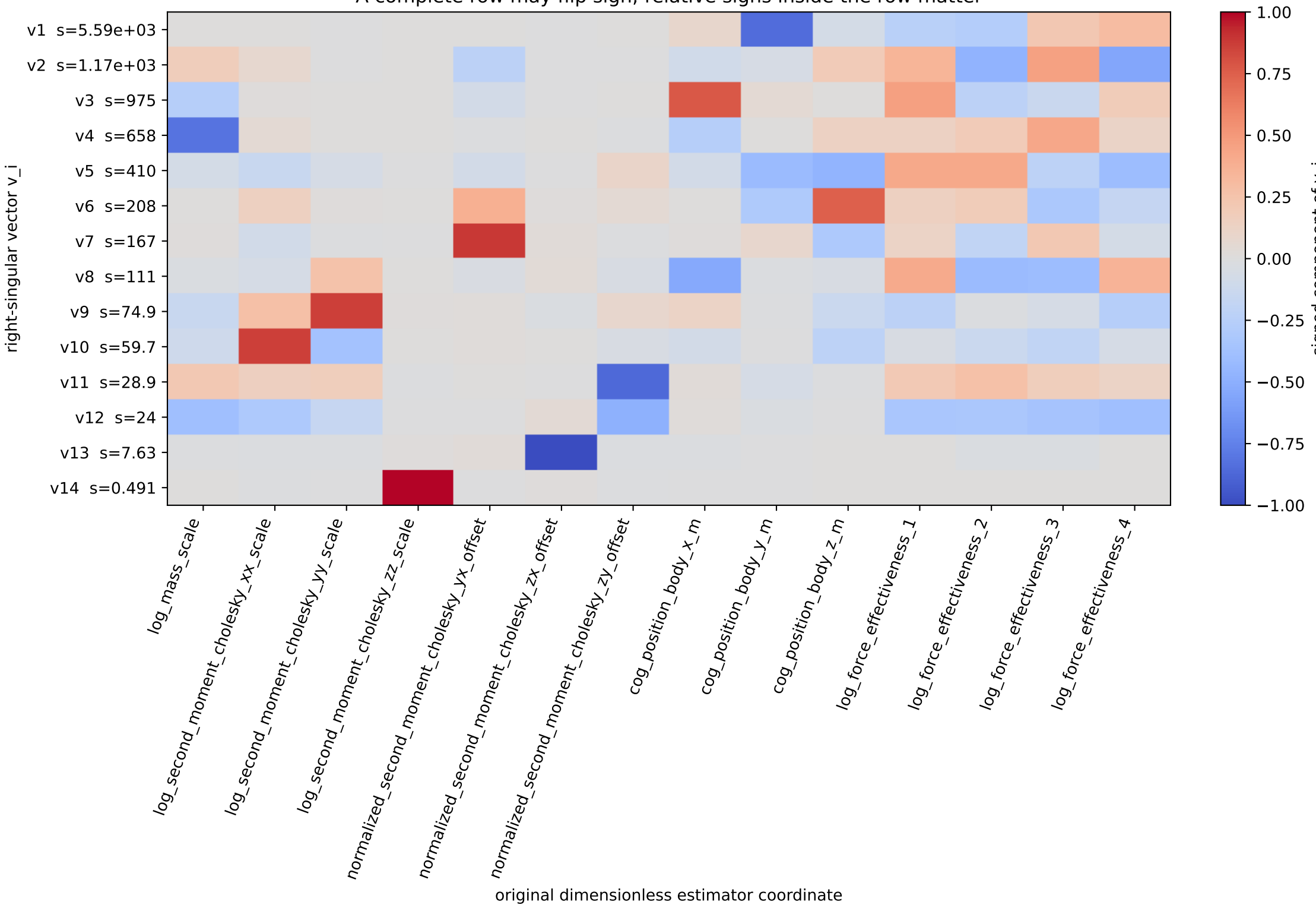
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36580814488

inertia J [kg m²]

[0.065010975 -2.5049121e-05 2.2553297e-05]
[-2.5049121e-05 0.065256059 7.2599536e-06]
[2.2553297e-05 7.2599536e-06 0.12899416]

principal moments [kg m²]

[0.065008433, 0.065258593, 0.12899416]

CoG position in body frame [m]

[-0.0023586288, 0.00086831555, 0.010529643]

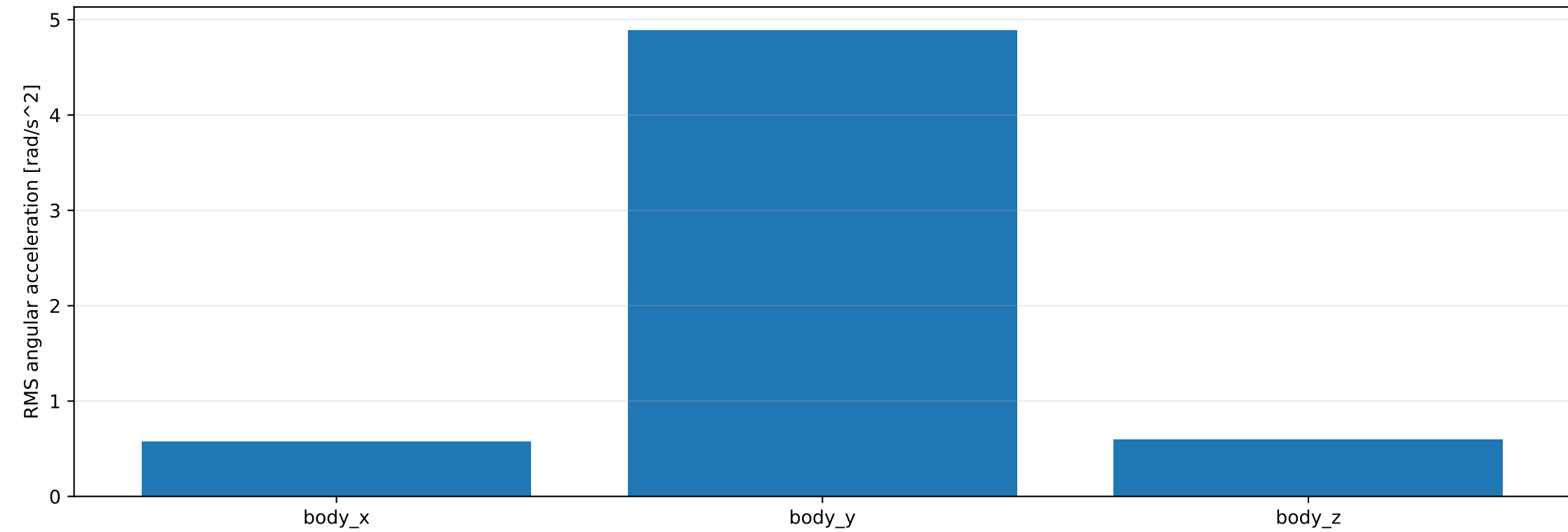
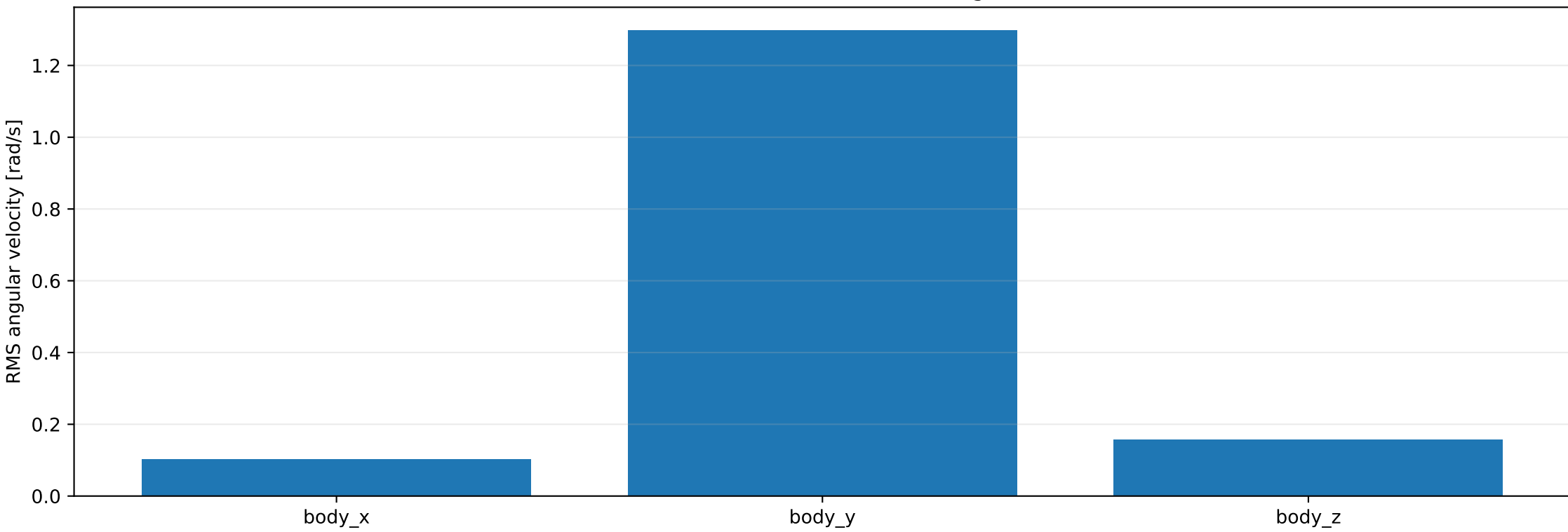
rotor force effectiveness

[0.96539781, 0.9731335, 0.95809751, 0.95487283]

selected delay [s]

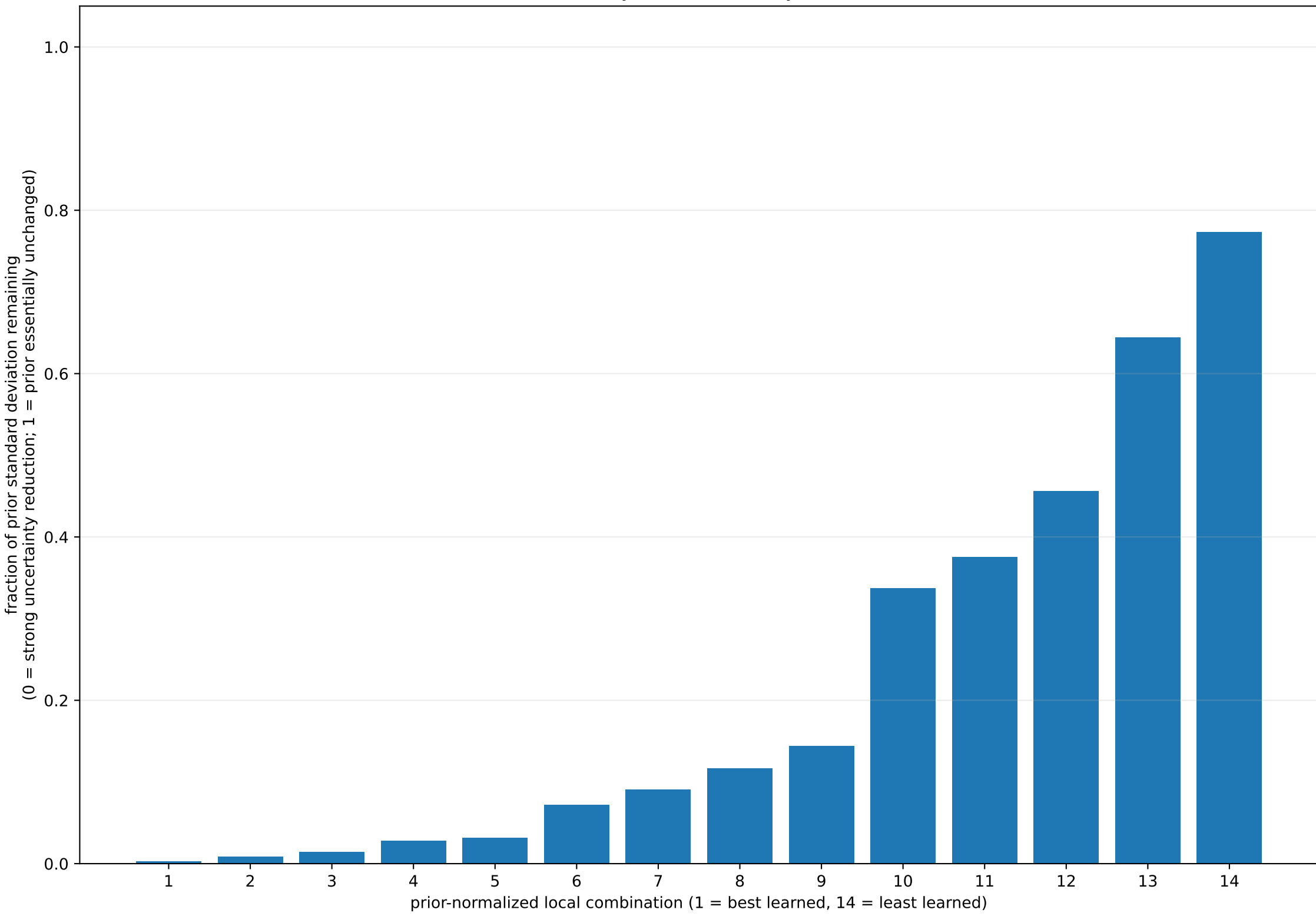
0.195002269745

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

